The International Journal of Advanced Manufacturing Technology
https://doi.org/10.1007/s00170-026-17896-y

ORIGINAL ARTICLE

Optimal sensor selection and press blow detection for hammer systems

Abdulkerim Karaman[1] · **Andreas Schwung**[2] · **Michael Marré**[1]

Received: 27 October 2025 / Accepted: 13 March 2026
© The Author(s) 2026

Abstract
One of the biggest challenges in the control of hammer systems is the reduction of the press blow and the avoidance of rebound impacts as it results in energy waste, tool wear and increased maintenance. In this paper, we take a step in this direction by proposing and evaluating various sensor systems and subsequent press blow detection systems which serve as a measure of the required forming energy. Specifically, we discuss and evaluate the potential sensor types, namely displacement/speed sensors, acceleration sensors as well as structure-borne sound sensors, as well as their optimal placement on the hammer system. Furthermore, based on the different sensor signals, we derive conditions for the press blow detection and intensity estimation within hammer systems. We evaluate the approach on a hammer system in a laboratory environment as well as in industrial serial production with very encouraging results indicating the strength and limitation of the different sensor systems.

Keywords Hammer systems · Press blow detection · Sensor selection and placement · Structure-borne sound sensors

1 Introduction

Machine hammer are forming machines used for forging [1]. Hammer systems are widely used both in series production, e.g. in the forging of crankshafts, and in individual part production in open-die forging of workpieces of all sizes and applied in various industrial domains. A characteristic of the mechanized forging hammer is that the forming tool transfers the deformation energy to the workpiece at high speed, in contrast to the rather slow movement in a press [2].

A general challenge in the operation of hammer system is that the forming energy required for forging of a workpiece is typically unknown and strongly dependent on various influencing variables including the forging operation, the used tool and it's actual condition, the workpiece material and potentially varying material characteristics [2, 3]. In addition, the state of forging, i.e. the estimation of the remaining forming energy required in case of multi-step forging, is notoriously difficult [4]. As a result, the forming energy is typically estimated before forging potentially resulting in additional, not required forging steps with corresponding press blows (unintended die contact). This strongly increases the wear of the tools in use [5] and consumes more energy than actually required. With constantly rising energy costs and the importance of sustainability of production processes, approaches to reduce the forming energy to required minimum is of utmost importance.

Although there are continuous setting hammer systems, which allow to better control the energy transfer within the hammer system, they are expensive and complex. Particularly, most hammer system in use are not continuously operated and often lack the required sensor systems. Consequently, hammer systems are operated solely by operator experience with the above challenges.

In this work, we aim to bridge this gap in that we tackle the problem of estimating the required forming energy of the last blow by reducing the press blow during hammer system operation. This shall lead to energy efficient control of the

✉ Abdulkerim Karaman
karaman.abdulkerim@fh-swf.de

Andreas Schwung
schwung.andreas@fh-swf.de

Michael Marré
marre.michael@fh-swf.de

[1] Labor für Massivumformung, South Westphalia University of Applied Sciences, Iserlohn, Germany

[2] Automation Technology and Learning Systems, South Westphalia University of Applied Sciences, Soest, Germany

Published online: 07 April 2026

forging process. To this end, we first identify suitable sensor systems to estimate the state of forging during operation. Particularly, we analyze three different systems including displacement/speed, acceleration as well as structure-borne (or acoustic emission) sound sensors regarding their ability to detect press blows and estimate their intensity. Second, we evaluate different sensor placement around the hammer system thereby deriving optimal sensor position. Third, we derive conditions indicating a press blow which hints on potentially unnecessary forging steps. Our key contributions can be summarized as follows:

1. We propose and analyse various sensor systems to measure detect and estimate the intensity of press blows serving as an indirect indicator of the required forming energy. We further derive optimal sensor location at the hammer to optimize the measurement's validity.
2. We derive a monitoring system for the press blow detection and press blow intensity estimation which allows the mitigation of press blows during operation.
3. We evaluate the system on a hydraulic double-acting hammer system both in a laboratory environment as well as in an industrial serial production with very encouraging results. Particularly, acceleration sensors allows both press blow detection as well as intensity estimation, paving the way for improved control performance.

This paper is organized as follows: We review related work in Sec. 2. The considered hammer system, the experimental setup, as well as the sensor systems, their optimal placement and the press blow detection methodology are presented in Sec. 3. The evaluation results are discussed in Sec. 4, and the conclusion is given in Sec. 5.

2 Related work

Hot hammer forging is process which is to date still mainly operated by user experience. Hence, the defining components of the hammer process are typically designed by trial-and- error rather than physical analysis [2]. Recently, various works aim to advance the hammer process by more detailed analysis of the process and changes to the process itself. In [6], affects on the stress–strain state and the sealing of axial defects is analyzed. The production of large workpieces with concave faces has been shown in [7]. A new method for plate forging is presented in [8] using V-shaped dies to increase the uniformity of the strain distribution [8]. Forging process improvements are also reported in [9]. In [10], a forging methodology has been proposed to manufacture thick-walled pipes without the use of a mandrel. This allows to reduce the deformation time and save energy consumption for heating the workpiece. While these approaches concentrate on process design, we address measurement and automation of hammer processes. In addition to empirically driven process design, numerical modeling and simulation approaches have been increasingly applied to hammer forging processes. Finite element based studies have been used to analyze the influence of process parameters on forging load, material flow and defect formation, often in combination with experimental validation [11]. Furthermore, numerical optimization techniques have been proposed to determine suitable energy and process parameter distributions in multi-blow hammer forging, enabling a systematic reduction of forging load compared to conventional trial-and-error approaches [12].

Another line of research considers the instrumentation of hammer systems with novel measurement systems. Vibration analysis has been performed in [13] combining a model and sensor based approach. Alternatively, high-speed cameras are employed to analyze the impact of velocity in a counterblow hammer [14] while [15] use a stereo-vision system for similar measurements. Moreover, a friction sensor for hammer system has been employed in [16] which allows to additionally measure the contact pressure during forging. Contrary, in this paper, we propose to use acceleration and structure-borne sound sensors for the analysis of the press blow as a measure for energy efficiency.

Reducing the energy of forging processes is an ongoing research field. The energy reduction for shafts forging has been discussed in [17]. An energy optimal model predictive control scheme has been proposed in [18]. In hammer-forging, some recent studies concentrate mainly on the energy consumption of the mechanical part ignoring the process-workpiece interaction [19, 20]. Other studies particularly concentrate on model based approaches for blow efficiency and energy consumption prediction [21]. Specifically, a spring-mass-damping model of machine and tool is combined with a process model to describe hammer-forming operations. A real-time capable model of hammer forging is presented in [4] where a surrogate model of the forging process is combined with a spring-mass-damping model which enables the prediction of blow efficiency and energy distribution after each blow with very high efficiency. The approach is further developed for real-time forging control [22].

However, a drawback of these approaches are the expensive derivations of accurate models by means of FEM simulations and surrogate models. Also, varying process conditions might be difficult to model. Hence, contrary to the model based approaches, we aim to derive the detection of press blow using a solely data based approach. To this end, we analyze different sensor types as well as their optimal placement regarding the detection of press blow as a measure for the required forming energy of the process.

3 Experimental laboratory set-up and results

In this section, we give an overview on the considered hammer system and state the problem.

3.1 Laboratory hammer system

Gravity-drop and additionally actuated hammers are forging machines, which raise a ram to a predefined height. Gravity together with potentially other energy sources accelerates the ram until it hits the workpiece and deforms it. Extreme conditions with respect to temperature, impact velocity and developed energy can be observed which makes the instrumentation of difficult. The considered belt drop hammer consists of a ram, anvil, foundation, frame, spring-damper system and the hydraulic system as schematically shown in Fig. 1.

The laboratory-level tests are carried out exclusively in the bulk forming laboratory. The belt drop hammer has a nominal energy of 8.8 kJ and a maximum ram speed of approximately 4.8 m/s. The desired energy or ram speed is adjusted by varying the ram's drop height. The tests are conducted using aluminum, EN AW-6082 (AlMgSi1), at room temperature using workpieces as shwon in Fig. 2. Since the belt drop hammer only allows for very limited repeatability of the drop height (exception: maximum drop height), the impact strength is influenced by varying the initial height of the feed material.

3.2 Instrumentation system

We investigate two types of sensor systems, namely a set of acceleration sensors as well as structure-borne sound sensors.

3.2.1 Acceleration sensor

To identify a suitable acceleration sensor, based on different load cases and a comparison of the acceleration/time profiles of the different sensors, the PCB Synotech M350B23 sensor type was selected as the primary sensor. This sensor is specifically designed for the shock loads encountered during impact, but has a limitation of a maximum operating temperature of 80°C, which must be taken into account during operational testing. Alternatively, the triaxial sensor PCB 356A01 is used, as its three axes allow it to be used in any orientation, while the uniaxial sensors are intended to be used in the direction of the load only.

The sensors are mounted using a thread on the sensor, adhesive surfaces, or, in the case of ring-shaped sensors, a central screw. Since inserting a thread into the machine elements or tools is not always possible, mounting using adhesive-bonded adapter plates was investigated as an alternative. The results of the acceleration measurements show no significant difference between the mounting variants with and without adapter plates.

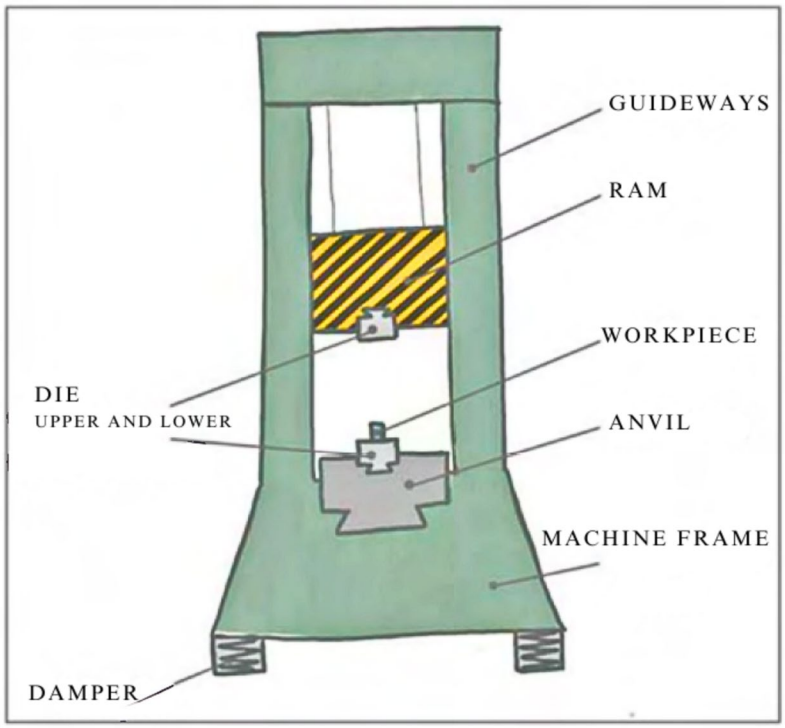


Fig. 1 Scheme of the considered belt drop hammer system in the bulk forming lab



Input material (⌀ 40.0)
Initial height H_0:
- 41.5 ± 0.2
- 43.4 +/- 0.2
- 54.6 +/- 0.2

Final geometry
- 1 forging blow
- Max. drop height
  - v_{max}: 4.8 m/s
  - E: 6.9 kJ

Fig. 2 Variants of the input workpiece material

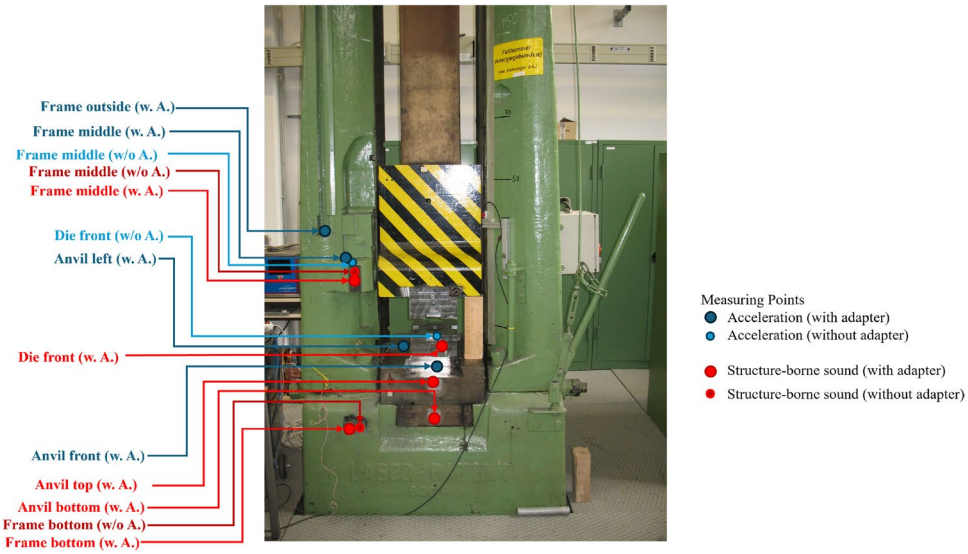


Fig. 3 Considered and evaluated sensor positioning at the belt drop hammer system

3.2.2 Structure-borne sound sensor

The structure-borne sound measurements are conducted using the Optimizer 4D measurement system and the O-WT 19 sensor are used all provided by QASS. In QASS's structure-borne sound analysis, these signals are recorded using high-frequency impulse measurement (HFIM) and converted into a three-dimensional image of the process using Fourier transformation. The graphical representation is based on the time, frequency, and amplitude axes. The advantage of this three-dimensional analysis is the significantly increased information content of the results.

The structure-borne sound sensors were mounted in the same way as the acceleration sensors.

3.3 Experimental results for acceleration sensors

Initial investigations were conducted to determine the required measurement rate for recording rebound impacts and to determine the influence of different positions (die, scraper) of the acceleration sensor. Figure 3 shows the investigated sensor positions. The influence of the position/location of the acceleration and structure-borne sound sensors on the forming unit or tool was investigated under three load cases, namely no impact, light impact and hard impact.

3.3.1 Experimental results

Figure 4 shows an example of the path, velocity, and acceleration curves for a forming stroke as a function of time. The times of initial contact between the die and the component and the onset of the impact were determined in the acceleration diagrams and then transferred to the displacement/velocity diagrams. The initial contact between the die and the workpiece, determined from the acceleration values, can be checked and confirmed by means of the initial and final height of the workpiece. The maximum acceleration values determined in the laboratory test for impacts were up to 8,000 g.

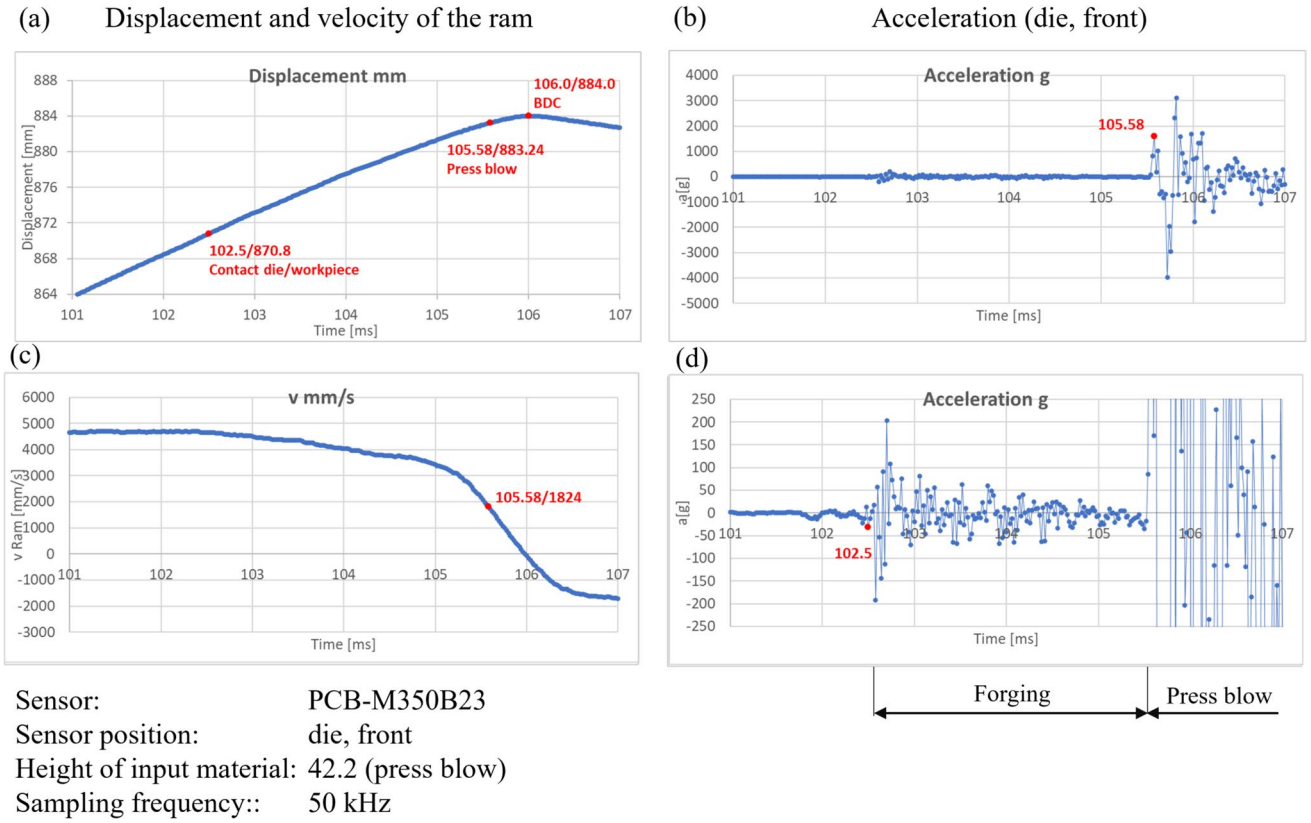
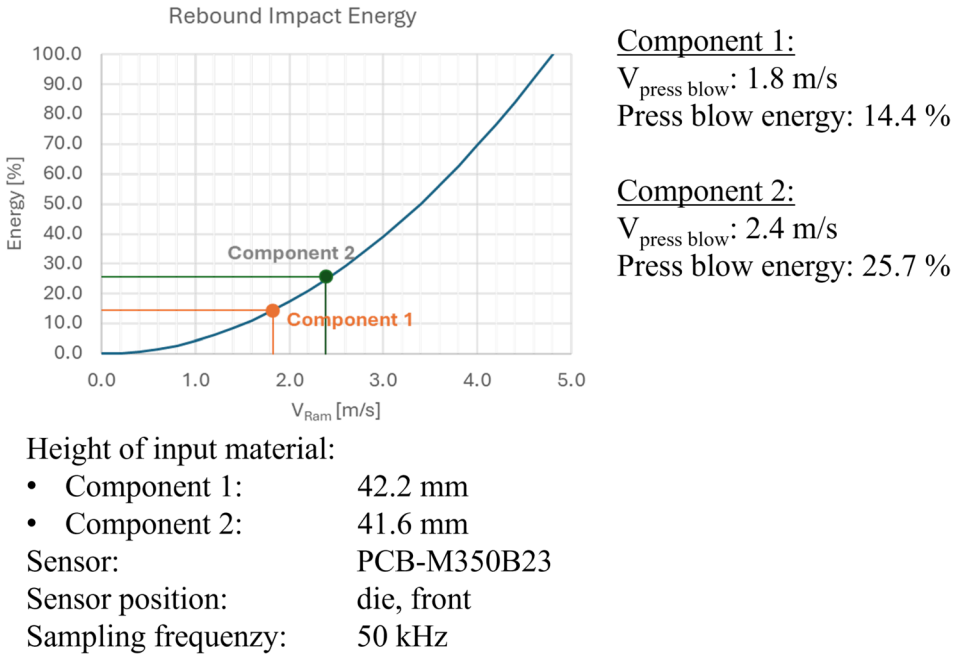


Fig. 4 Example acceleration measurement for one forming impact (with press blow impact).(a) Ram displacement as a function of time.(b) Acceleration signal as a function of time.(c) Ram velocity as a function of time.(d) Zoomed-in acceleration signal as a function of time

Fig. 5 Share of impact energy



We further analyse the proportion of rebound impact energy relative to the total kinetic energy in Fig. 5. The total energy (100%) is determined by the impact velocity of the upper die/component and corresponds to the maximum ram speed for the belt drop hammer used. The example shown demonstrates the high sensitivity of rebound impact energy

to process variations. A reduction in the thickness of the feed material by 0.6 mm leads to a relative increase in rebound impact energy of more than 50%.

3.3.2 Sensor placement

To compare and evaluate the sensitivity of the acceleration measurements with respect to the intensity of the impacts and the position of the sensors, the acceleration measurement results are normalized to the “no impact” case. Figure 6 shows that at the sensor position "center of frame," there is only a slight difference in the acceleration behavior depending on the impact intensity. In comparison, at the sensor positions "front of frame" and "front of die," a impact can be reliably identified, regardless of the intensity.

3.4 Experimental results for structure-borne sound sensors

We start the discussion with the graphical representation of the structure-borne sound measurement results as shown in Fig. 7. Obviously, different amplitude values depending on the impact intensity are clearly shown. However, while the 3D graphical representation is very informative, it does not allow to directly quantify the results, nor can it be used as a control variable for process control. For this reason, we introduce the “sound volume” as a quantitative measurement. The sound volume P_{sv} is calculated from the product of average amplitude $\bar{A}$, time t_{pb} of the press blow, and a pre-defined frequency range Δf as

$$P_{sv} = \bar{A} \cdot \Delta \cdot t_{pb} \tag{1}$$

Thus, a measured value is available as the result of the impact intensity.

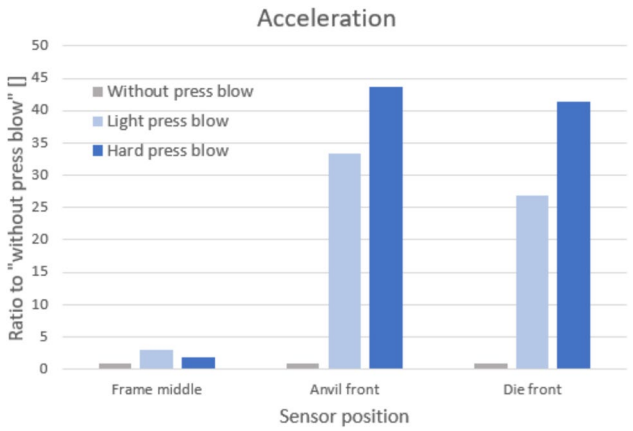


Fig. 6 Standardized representation of the acceleration measurement

To further compare and evaluate the sensitivity of the structure-borne sound measurements with respect to the intensity of the impacts and the position of the sensors, the results of the structure-borne sound measurements are standardized to the “no impact” case, analogous to the acceleration measurements. Figure 8 shows, that similar to the acceleration measurements, at the “center frame” sensor position, there is only a slight difference in the sound volume depending on the impact intensity. The greatest differences are achieved at the “bottom frame” sensor position, followed by the “bottom scraper” and “front die” sensor positions.

3.5 Discussion of experimental laboratory results

Both measurement methods enable a structured and sensitive identification of press blows, and in both cases, the results depend significantly on the position of the sensor. Also, it is worth noting, that not only the detection of press blow is possible, but the intensity differences between “light” and “hard” impacts are also recorded by both methods. In summary, the laboratory tests show that both acceleration measurement and structure-borne sound measurement appear suitable for the purpose of press blow detection. Hence, motivated by the promising laboratory assessments and to obtain a more thorough and concise analysis, we expand our experiments within an industrial serial production environment as presented next.

4 Experimental results for serial production in an industrial environment

In this section, we report the experimental results of the presented sensor selection and press blow detection within an industrial environment. To this end, we first discuss the serial production setup followed by the results obtained.

4.1 Experimental setup

We start by providing the experimental setup. The tests at the factory level are conducted under series production conditions. The forged component is a connecting rod (double piece) operated with a heating temperature $> 1200\,^{\circ}\mathrm{C}$ The impact sequence consist of 5 blows, with the first blow being the flattening followed by 3 preforms blows and a final forming blow. The considered forming unit is a double-acting hammer, fully automated Lasco HO-U 630 hammer. For the initial operational test, acceleration and structure-borne sound sensors were applied to various positions on the downstroke hammer as shown in Fig. 9. For acceleration measurements, we use two different sensor types, namely the uniaxial sensor PCB M350B23 and the triaxial sensor PCB 356A01. The uniaxial acceleration sensor was used in four

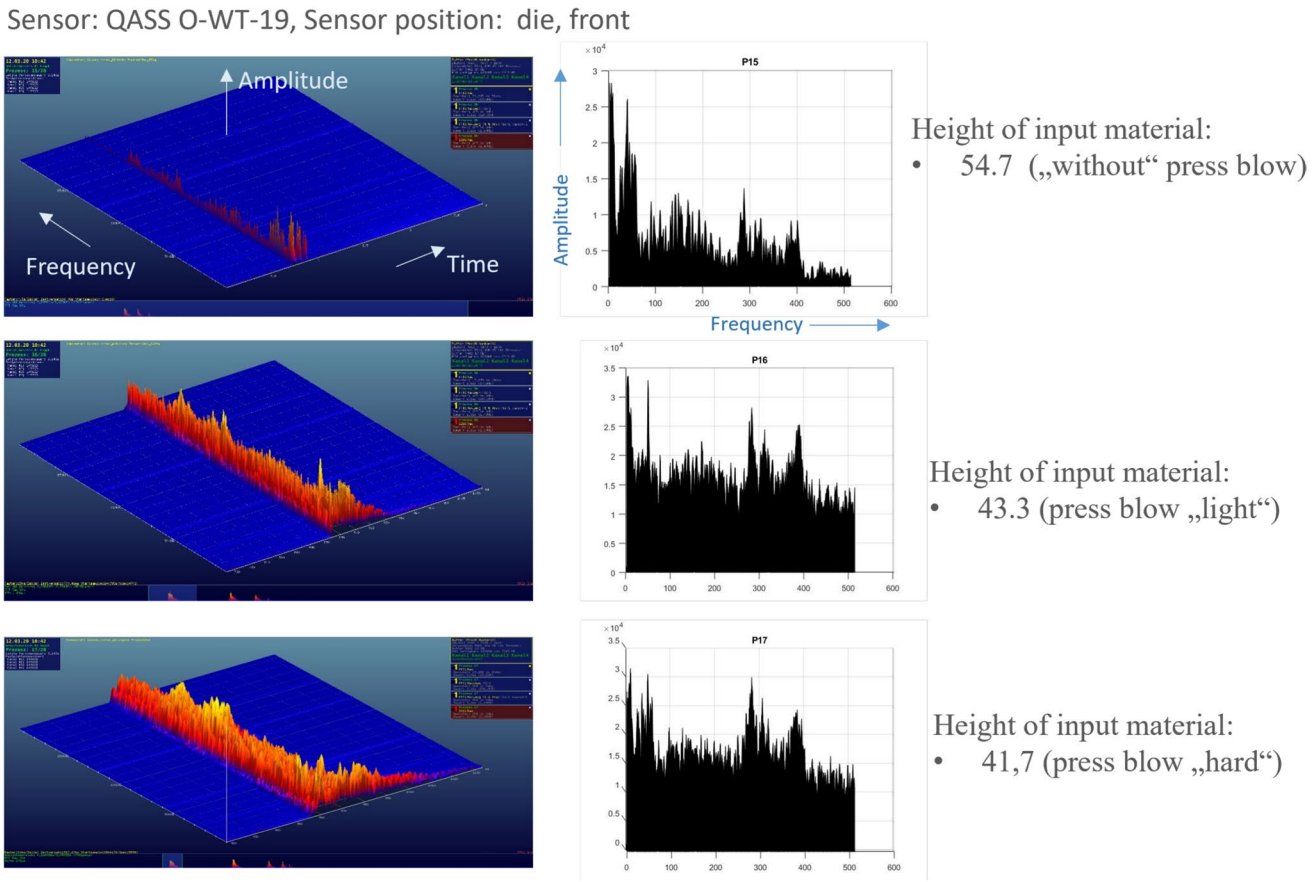


Fig. 7 Results of the structure-borne sound measurement as a function of the impact intensity

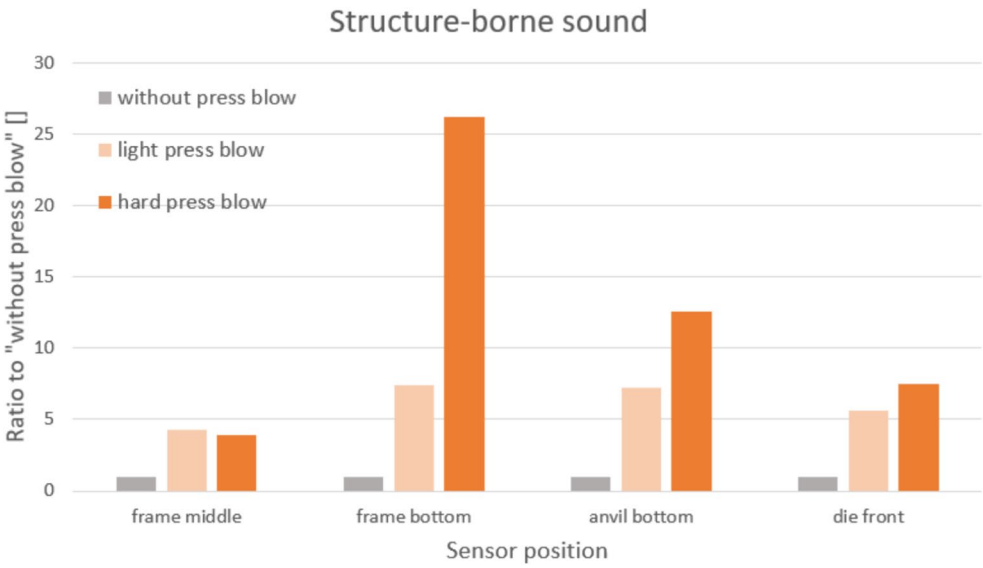


Fig. 8 Standardized representation of the structure-borne sound measurement

different positions, while the triaxial sensor, due to its limited measuring range of +/- 1,000 g, was used exclusively in the “frame down” position. Both sensors are used in parallel. Furthermore, as a third alternative, a displacement sensor was also used to record the ram’s paths and speeds.

To estimate the expected thermal load on the sensors before the start of the test, the temperature profile in the heat-critical system areas (scrapes and frame next to the scrapes) was continuously measured during production over a period of more than 24 hours. The results show that

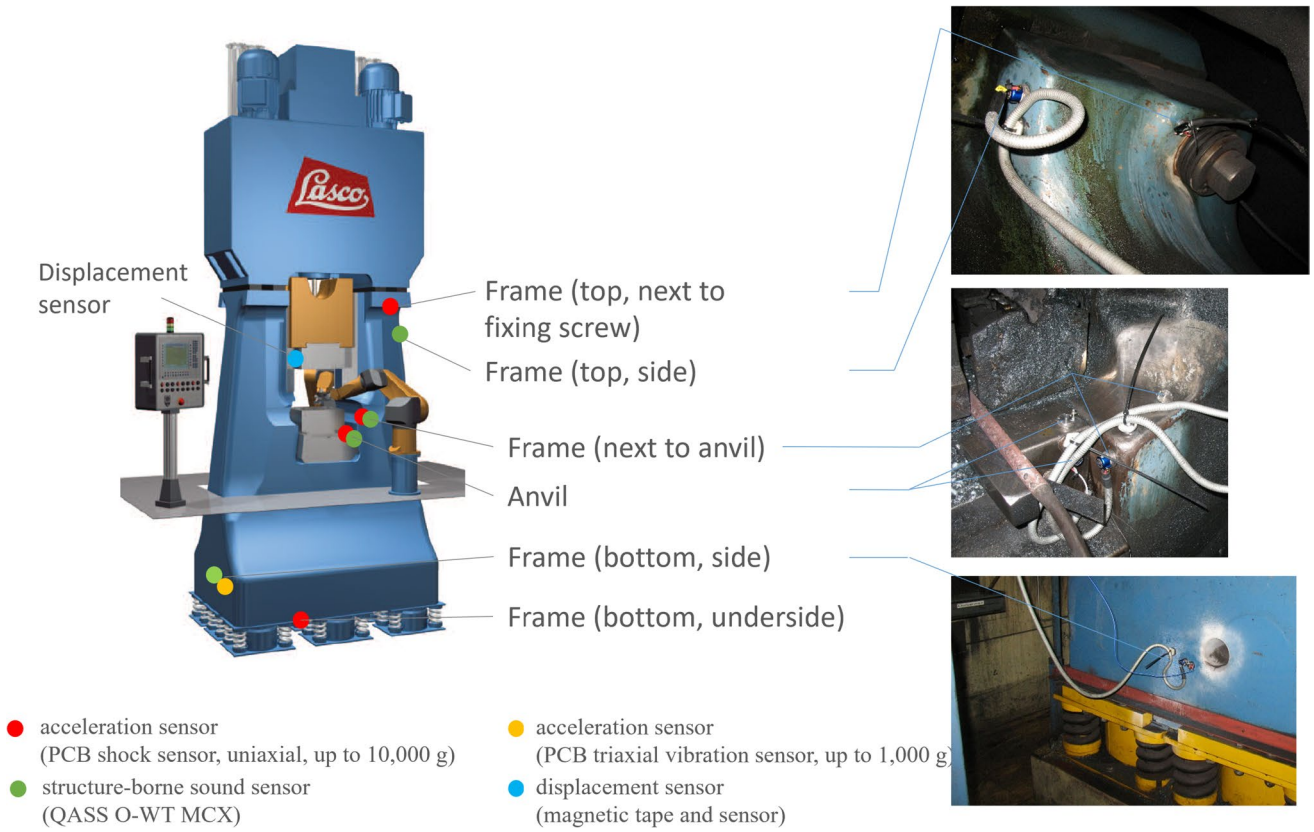


Fig. 9 Standardized representation of the acceleration measurement

the temperature limit of 800°C for the intended use of the accelerometer is not exceeded. Furthermore, during the long-term measurement the application of the sensors using glued adapter plates could be validated successfully.

4.2 Results

We present the results in three steps. First, we conduct a signal analysis to identify the different stages of the press blow. Subsequently, we propose two criteria for press blow detection based on acceleration measurement. To further enhance the signal analysis, we propose an energy related press blow intensity evaluation to allow for more detailed downstream operation to be used in the future. Lastly, we evaluate the structure-borne sound sensor regarding press blow detection capabilities.

4.2.1 Sensor validation and signal analysis

The basic path and speed of the ram during a forming stroke are shown in Fig. 10 using the example of the 5th stroke of a forming operation. To this end, we define the 1st contact between upper die and component as s_{contact}, the contact of the rebound surfaces, beginning of rebound as s_{rebound}, the bottom dead center as s_{BDC} and the point $s = 0$ as the upper/lower dies rest on each other (rebound surfaces). Also we have maximum speed of the ram v_{max}, the impact speed at upper die/component as v_{contact} and at rebound surfaces of upper/lower die as v_{rebound}, the speed at BDC as v_{BDC}, the maximum return speed as v_{ret} and the maximum upward speed as v_{min}. The diagrams indicate, that the forming time (time between s_{contact} and s_{BDC}) is approximately 1 ms, and that the acceleration of the ram during the return stroke approximately corresponds to the negative acceleration during forming as the gradient of the speed curve remains almost identical immediately before and after BDC.

4.2.2 Press blow analysis using path and speed

The term "press blow" generally refers to the collision of the upper and lower dies with the designated tool surfaces, i.e. the contact surfaces. To define and clearly detect rebound impact, we propose a combination of two criteria, namely the travel and springback of the ram. More specifically, we define two conditions for the rebound impact to occur: First, the travel of the ram must be below BDC, i.e., the dies are slightly elastically deformed yielding

$$s_{\mathrm{BDC}} > 0. \tag{2}$$

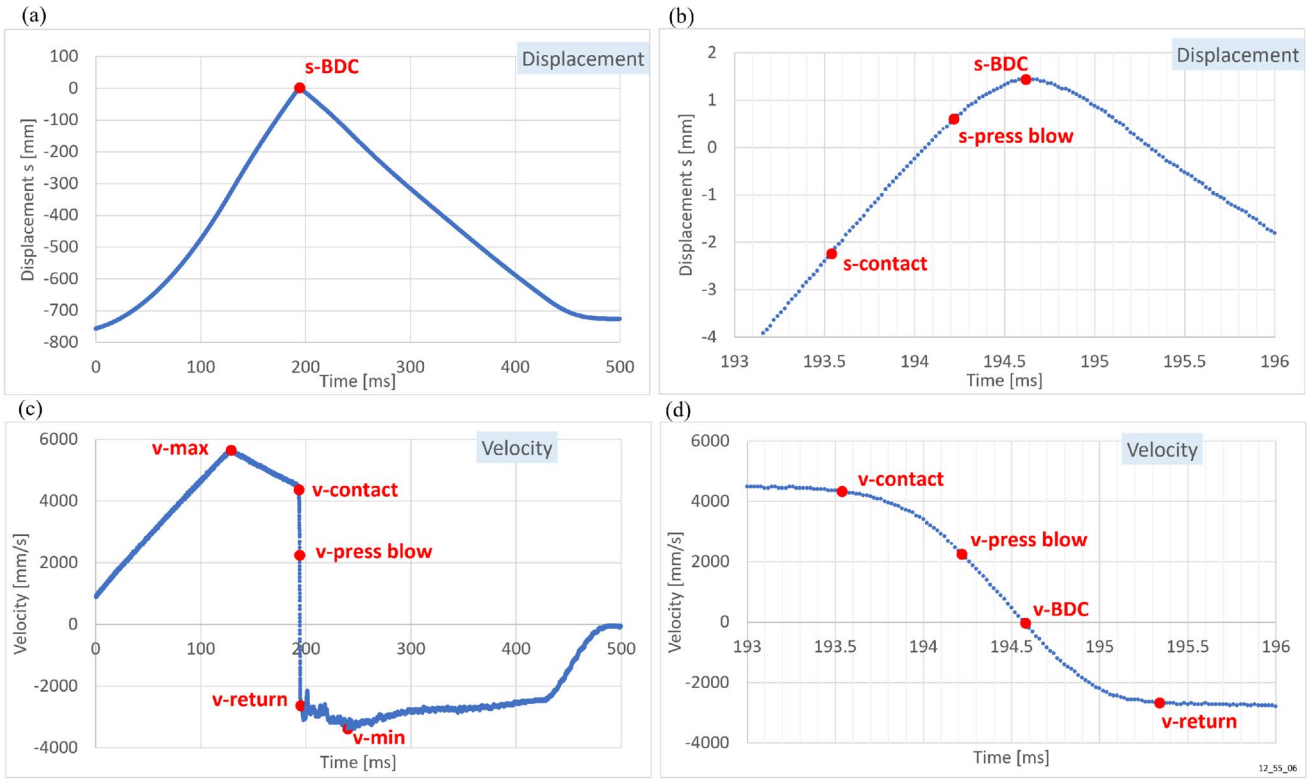


Fig. 10 Path and speed of the ram at 5*th* blow. (a) Displacement as a function of time for the complete stroke. (b) Zoomed displacement around die–workpiece contact, press blow and bottom dead center (BDC). (c) Velocity as a function of time for the complete stroke. (d) Zoomed velocity around die–workpiece contact, press blow and bottom dead center (BDC)

Second, the ratio of the impact velocity v_{contact} to the maximum rebound velocity v_{rebound} must be greater than a certain negative treshold r_{tres}, i.e.

$$\frac{v_{\text{contact}}}{v_{\text{rebound}}} > r_{tres}. \tag{3}$$

We illustrate the first and second condition in Figs. 11 and 12, respectively.

For the first criterion, the ram's paths are listed in Fig. 11 for all 5 strokes in a sequence. It can be seen that Criterion (2) is met for the 4*th* and 5*th* strokes which is in line with the observations of press blow during production. The second criterion (3) confirms the observation as shown in Fig. 12. Here, Criterion (3) is met for e.g. $r_{tres} = -2.5$. We note, that the approach relies on the determination of the ratio treshold which is potentially varying and has to be adjusted accordingly.

The measurement results of the first operational test with a 12-fold loading for displacement and speed measurements are listed in Fig. 13. The difference between the first three blows and the last two are clearly visible, with the last two having positive BDC as shown in the upper right figure. We note, that the diagram shows a repeatability of $+/-0.1mm$ for the 4th and 5th impacts. Furthermore, Fig. 13 shows the maximum ram speeds for each impact in the lower left diagram, which are directly proportional to the maximum kinetic energy (lower right diagram). The diagram shows that the energy increases relatively slightly from impact to impact during the first three impacts and then reaches significantly higher values for the press blow in the 4th and 5th impacts.

The rebound velocity (impact velocity of the rebound impact surfaces of the upper/lower die) and the rebound velocity are characteristics for identifying rebound impacts. To evaluate and compare the intensity of rebound impacts, we normalize the rebound velocity of the 4th and 5th impacts to the 3rd impact (no rebound impact) which serve as an additional criterion for press blow detection. As we are also interested in the press blow intensity we also report the ratio of the speeds of the 4th and 5th impacts to each other. In theory, the 5th impact should have higher intensity than the 4th impact. The lower diagrams in Fig. 14 show that the rebound velocity of the 4th and 5th impacts is approximately a factor of 4 greater than that of the 3rd impact, but a significant difference in intensity between the 4th and 5th impacts is not discernible.

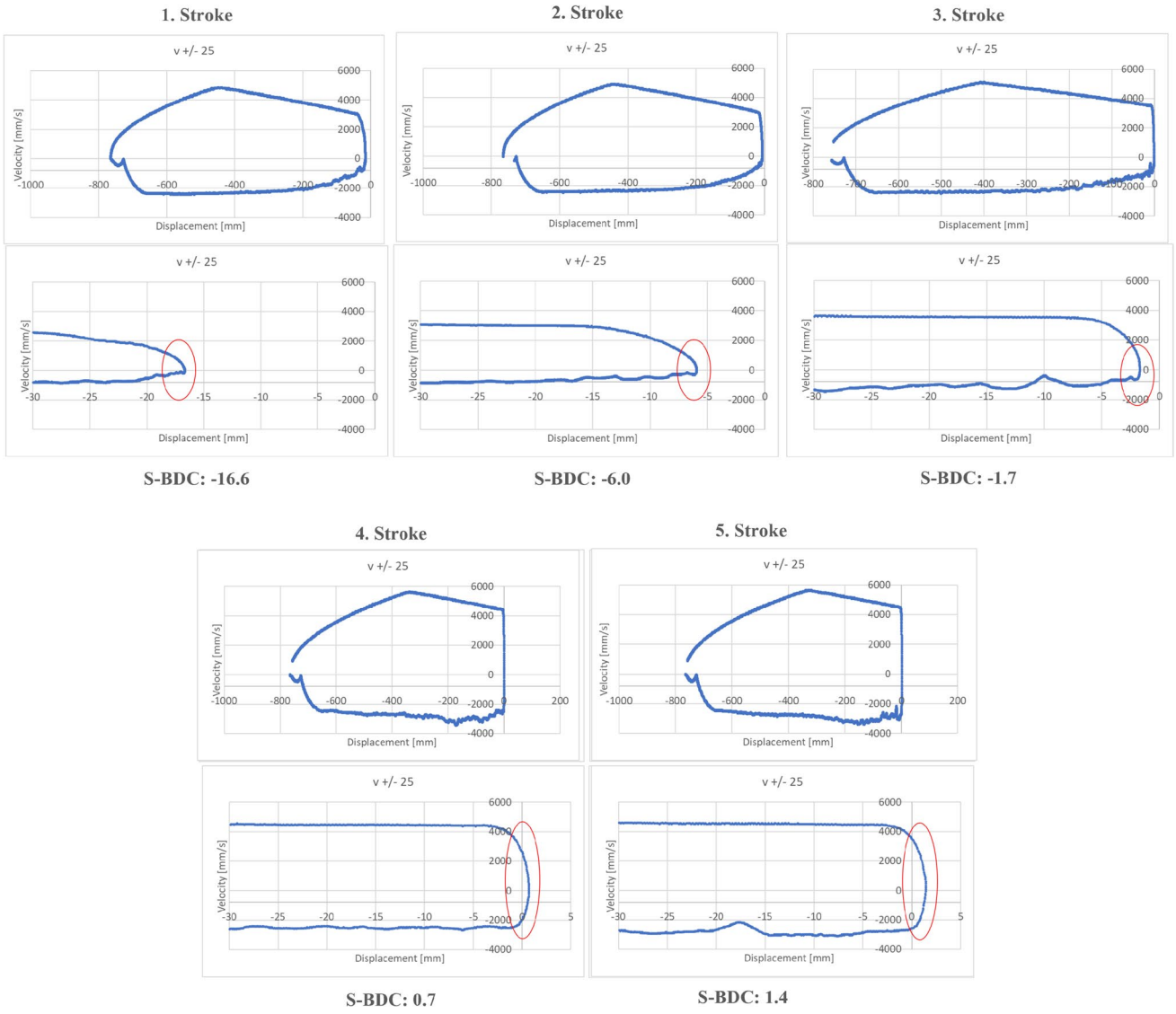


Fig. 11 Press blow criterium BDC

Lastly, based on the speed measurement and the ability to determine the first contact point between the upper die and the component, the energy components for each individual impact can be calculated. The maximum kinetic energy of the ram

$$E_{\max} = \frac{1}{2} m \cdot v_{\max}^2. \tag{4}$$

is divided into the components braking energy, forming energy, and rebound impact energy as follows:

$$E_{\text{brake}} = \frac{1}{2} m \cdot (v_{\max}^2 - v_{\text{contact}}^2), \tag{5}$$

$$E_{\text{forming}} = \frac{1}{2} m \cdot (v_{\text{contact}}^2 - v_{\text{bounce}}^2), \tag{6}$$

$$E_{\text{rebound}} = \frac{1}{2} m \cdot v_{\text{bounce}}^2, \tag{7}$$

Note, that a part of the rebound energy also contributes to the forming process.

4.2.3 Press blow analysis using acceleration sensors

The accelerations during forming were recorded using the above mentioned two sensors types.

We first present the results of the uniaxial acceleration sensor at different positions in Fig. 17. Depending on the sensor position, clear differences in the measured values of a blow sequence can be observed. The lowest values are achieved at the “frame bottom” position. Here, almost no reaction is measurable for the first and second blows. The

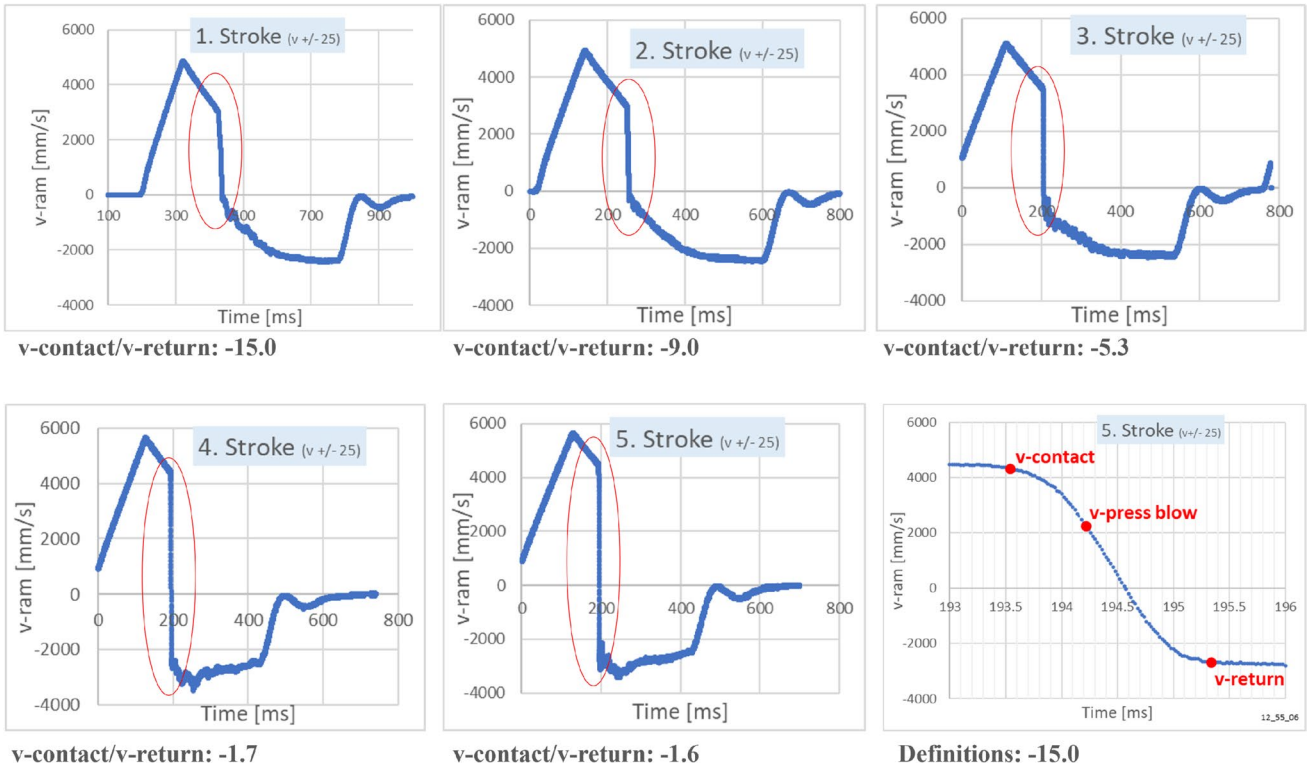


Fig. 12 Press blow criterium springback

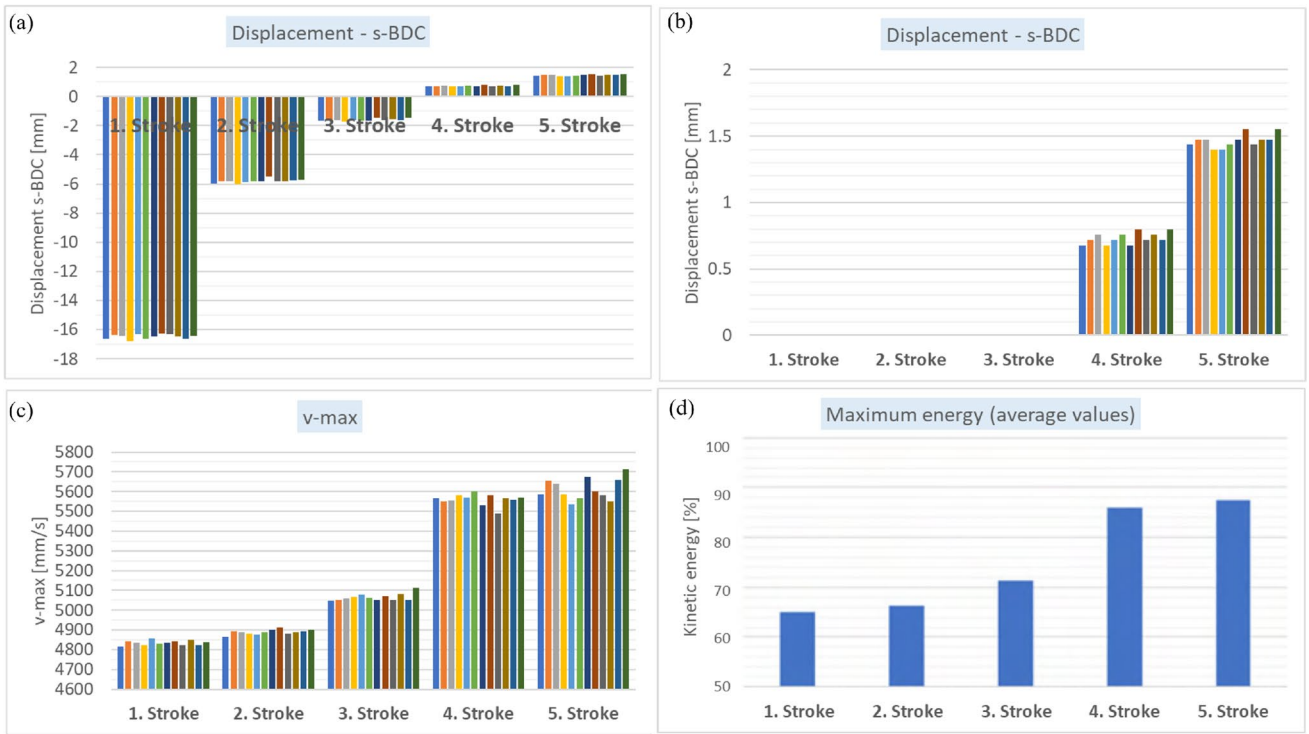


Fig. 13 Measurement results for a 12-fold loading sequence. (a) Bottom dead center (s-BDC) position for each stroke. (b) Zoomed view of the s-BDC position showing a repeatability of ±0.1 mm for the fourth and fifth stroke. (c) Maximum ram velocity for each stroke. (d) Mean maximum kinetic energy for each stroke

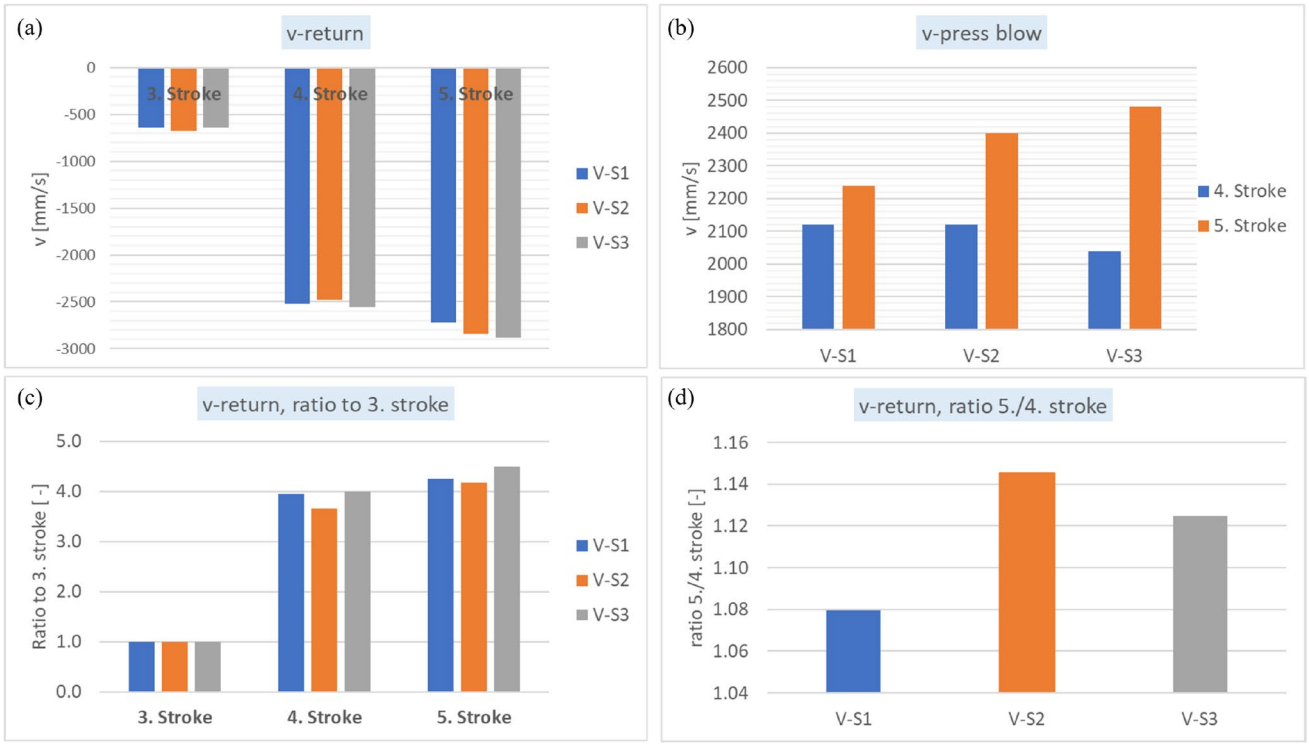


Fig. 14 Measurement results for distance (upper figures) and speed (lower left) as well as the calculated mean of maximum energy (lower right) for a 12-fold loading. (a) Distance for all strokes. (b) Zoomed distance for the fourth and fifth stroke. (c) Speed for all strokes. (d) Mean of maximum energy

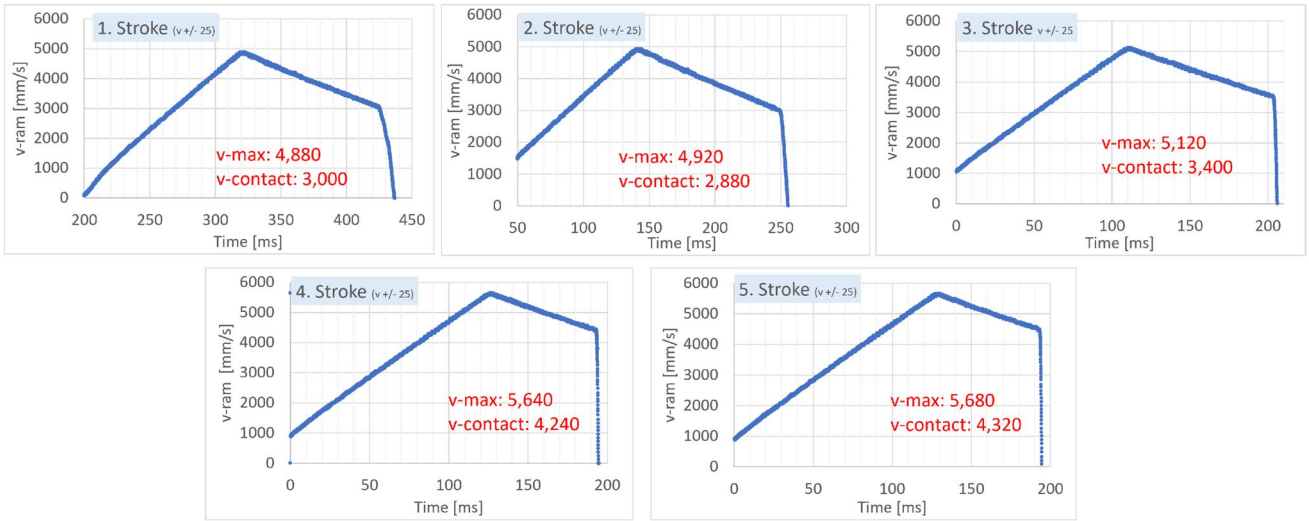


Fig. 15 Speed curve for all strokes in a stroke sequence

reason is likely to be the large, compact mass of the hammer frame, which has a strong damping effect on the acceleration. By far the highest acceleration values are measured at the "scraping" position. The results from this sensor position are detailed in Fig. 18. Here, not only the significant differences in the acceleration values depending on the blow sequence are visible, but also, using the example of the fifth blow, the time range of the forming process and the onset of the rebound impact.

An overview of the minimum and maximum acceleration values depending on the impact sequence and sensor position is shown in Appendix Fig. 23 with the normalization procedure as discussed before. The normalized representation shows that even a sensor position with low absolute

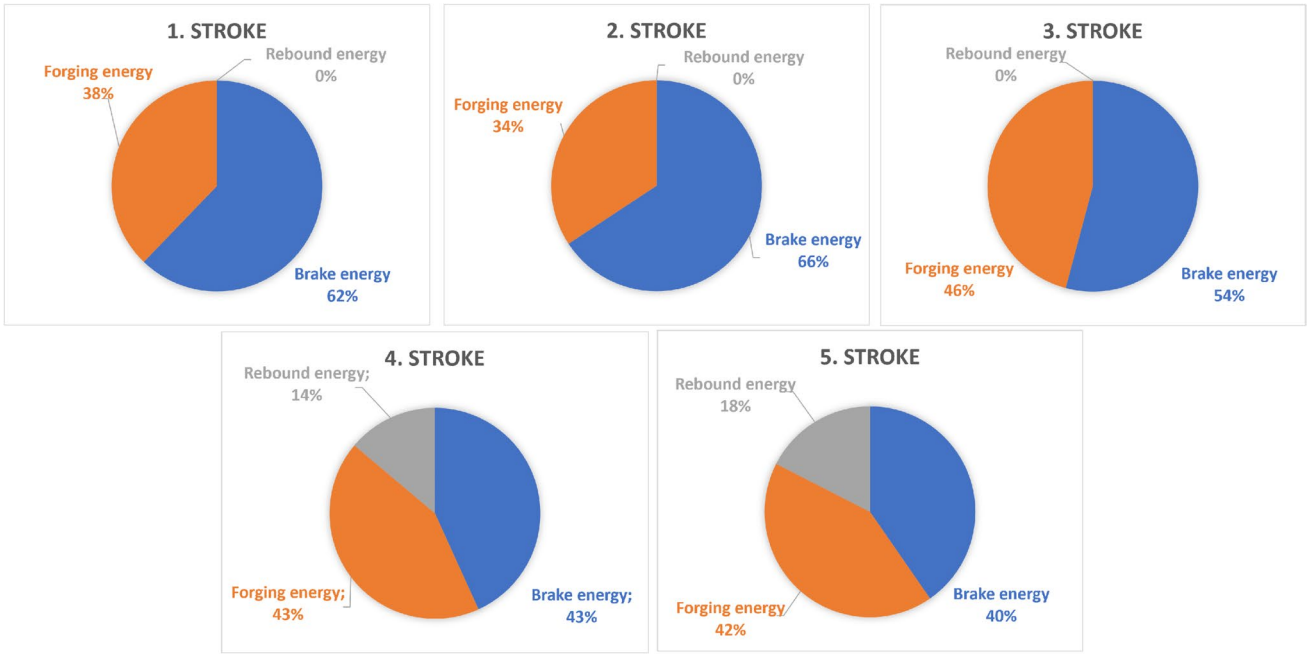


Fig. 16 Distribution of energy for all strokes in a sequence

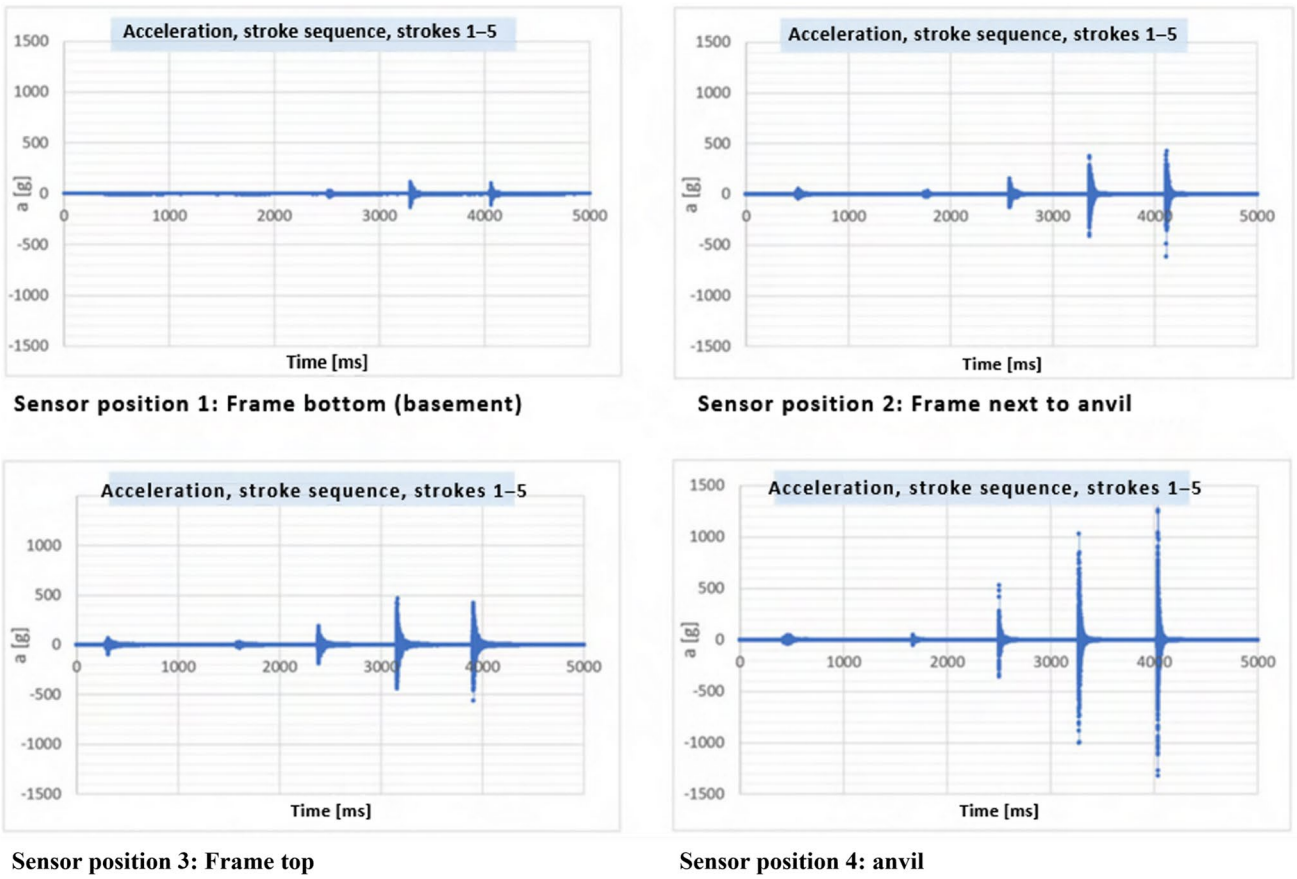


Fig. 17 Acceleration measurement results using the uniaxial acceleration sensor for five impacts

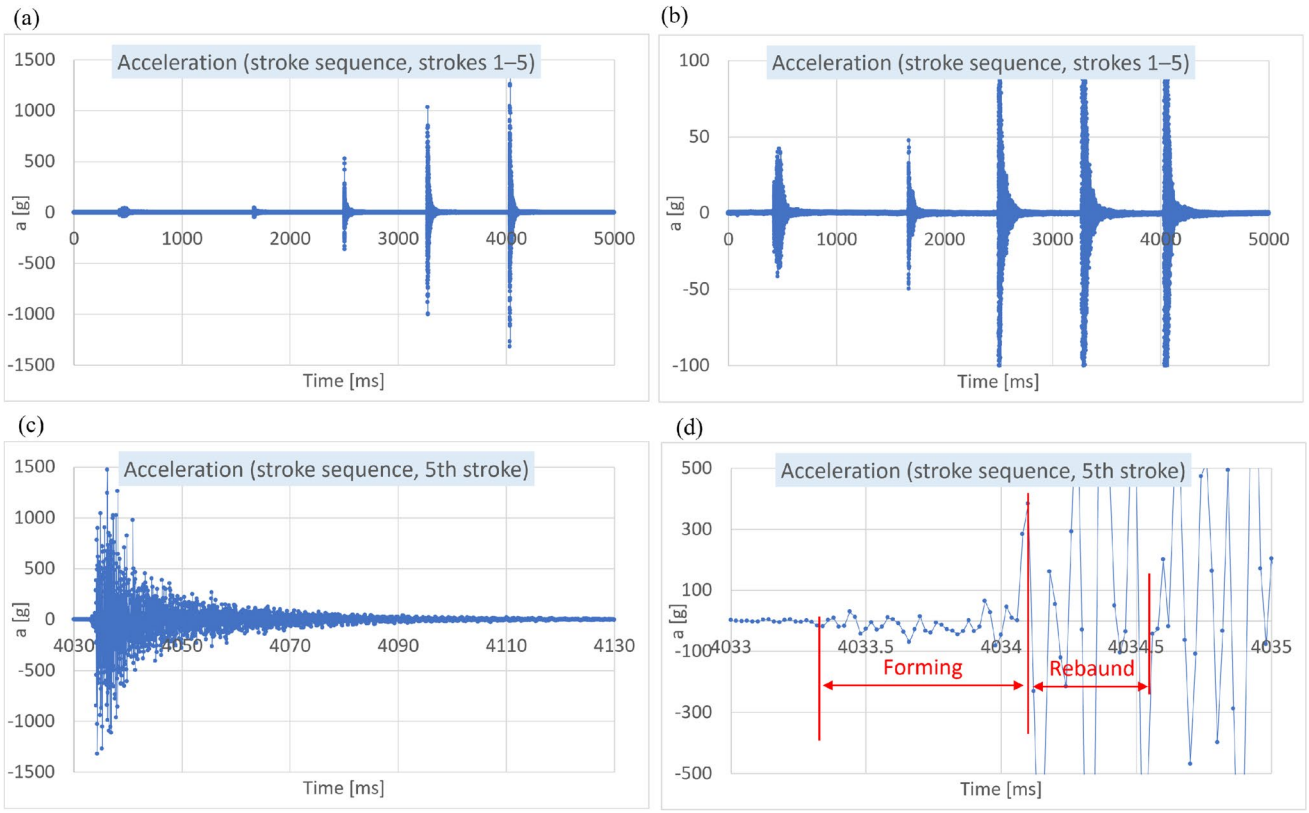


Fig. 18 Acceleration measurement results using the uniaxial acceleration sensor for the fifth impact. (a) Acceleration signal for the complete stroke sequence (strokes 1–5). (b) Acceleration signal for the same stroke sequence with adjusted scale. (c) Acceleration signal of the fifth impact. (d) Zoomed view of the fifth impact highlighting the forming and rebound phases

acceleration values, such as the “frame down” position, enables the rebound impacts (impacts 4 and 5). Contradictory results are visible for the ratio of the acceleration of the 5th to the 4th impact depending on the sensor position. For the positions close to the die (scraper, frame next to the scramble), ratios significantly greater than 1.0 are achieved, which means that the 5th impact has higher accelerations than the 4th impact. For the positions farther from the die (frame top/bottom), most ratio values are near to 1.0. Here, the statement regarding the intensity of the rebound impacts is questionable.

The measurement results with the triaxial sensor demonstrate the increased sensitivity of the sensor compared to the uniaxial sensor. Although the sensor was mounted far from the die on the "frame, bottom, side," all impacts of a sequence of impacts are visible and the rebound impacts are identifiable. These results are confirmed by further evaluation and normalization of the measured values, see Appendix Fig. 24. However, the results regarding the intensity of the rebound impacts, determined by the ratio of the measured values of impacts 5 and 4, are inconsistent and questionable depending on the three measurement axes. This can be attributed to the required sensor position.

4.2.4 Press blow analysis using structure-borne sound sensors

The structure-borne sound measurements during forming were conducted in parallel with the acceleration measurements using the uniaxial sensor at four different positions on the forming machine. The measurement results allow the identification of the five impacts in a sequence of impacts, regardless of the sensor position.

To allow a quantitative comparison of the measurement results we propose to use the signal energy, i.e. the energy within a rectangular window (in time and frequency), which is successively shifted in time as shown in Fig. 19. The time and frequency range under consideration can be selected depending on the requirements. The signal energy essentially corresponds to the sound volume introduced in the laboratory tests and provides a measured value for the intensity of a press blow, which allows for a quantitative comparison of results or could be used as a control variable for process control.

The signal energy results normalized to the third impact in Fig. 20 left show that the rebound impacts are identifiable at all sensor positions. The best results are achieved with the "frame next to the scabbard" position, and the worst with the

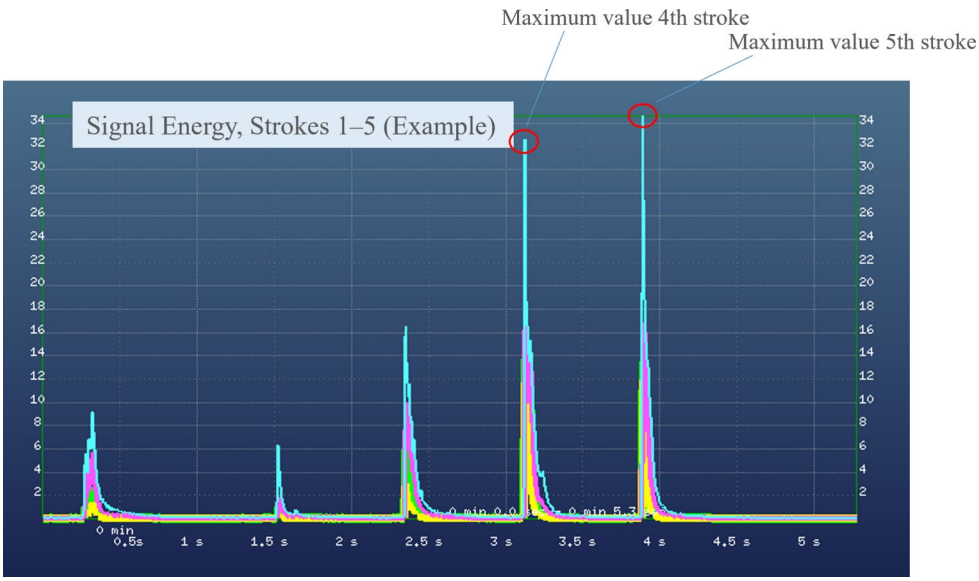


Fig. 19 Signal energy for an impact sequence

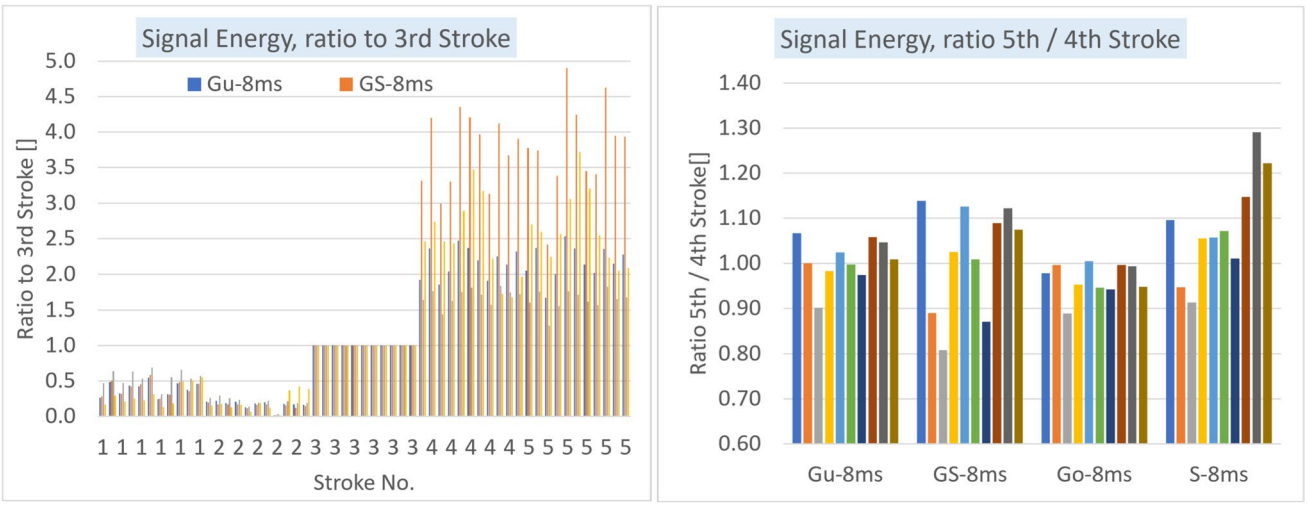


Fig. 20 Normalized structure-borne sound measurement results for a 10-fold loading: ratio to the 3rd stroke (left) and ratio of the 5th to the 4th stroke (right). Position of the structure-borne sound sensor: S: canopy; GS: frame next to canopy; Go: frame above; Gu: frame below. Signal energy parameters: time window 8 ms, full frequency range

"frame on top" position. Again, we analyze the sensor capabilities regarding the press blow intensity by a comparison of the 4th and the 5th blow. The results in Fig. 20 right show that the intensity of the bounce impacts, determined by the ratio of the measured values from impacts 5 and 4, are not clearly represented for any sensor position.

5 Discussion

In this section, we discuss the obtained results in terms of press blow detection and intensity estimation. Further, we present an analysis of the implications of the findings in terms of the sustainable forging operations.

In general the sensors used for all measured variables (displacement, acceleration, structure-borne sound) showed no malfunction throughout the entire period of use, both in laboratory and operational tests. Long-term stability can only be confirmed for the displacement measurement, as the system used has also been in continuous operational use for many years. The measurement system for highly dynamic recording and analysis of measured variables proved suitable even in the harsh environment of the operational tests. Compared to model-based approaches relying on FEM or surrogate models [4, 21, 22], the sensor-based methods offer a cost-effective and transferable alternative. Transferability to other hammer types, such as counterblow hammers and screw presses, appears realistic.

5.1 Press blow detection

All three measurement methods allow for the clear identification of press blows and rebound impacts as shown in Fig: 21. For acceleration and structure-borne sound measurements, the results depend significantly on the position of the sensor. When optimal positions are selected, all three methods show comparable detection performance. In addition, both acceleration and acoustic emission signals are able to distinguish between light and hard impacts.

5.2 Press blow intensity estimation

The 5th impact exhibits a higher rebound velocity/energy than the 4th impact. Consequently, an intensity estimation is expected to indicate a higher intensity in the 5th impact in relation to the 4th impact. This intensity differences are best captured by the uniaxial acceleration measurement method with the sensor positioned at the canopy as compared in Fig. 22. The other two measurement methods however are less suitable for capturing rebound intensity from the signal energy. However, we

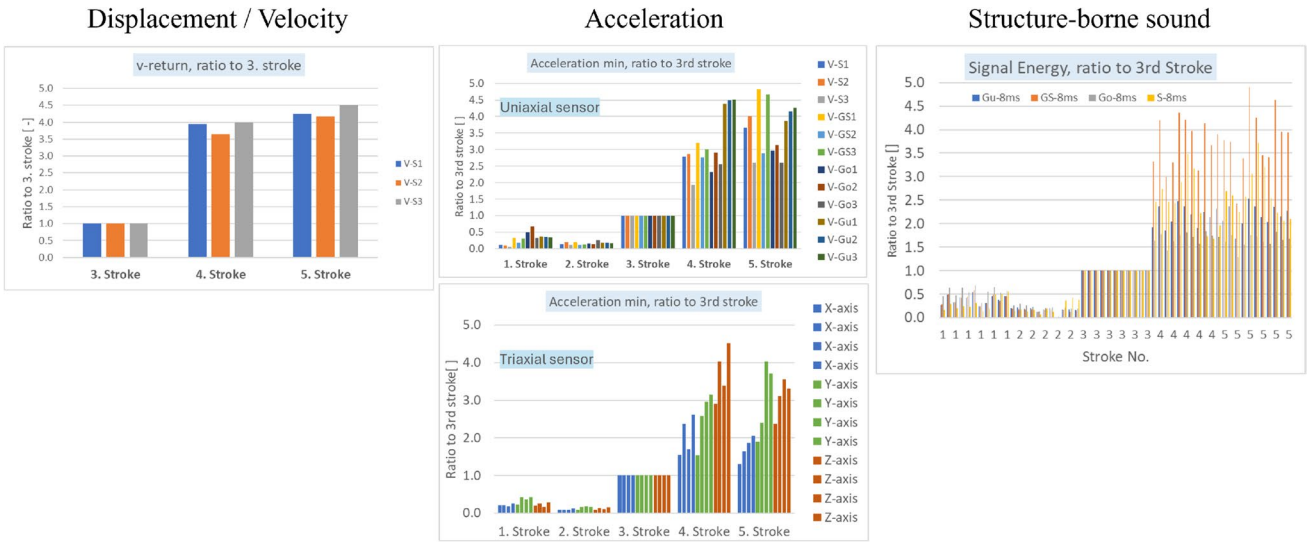


Fig. 21 Overview of the results on the problem of press blow detection

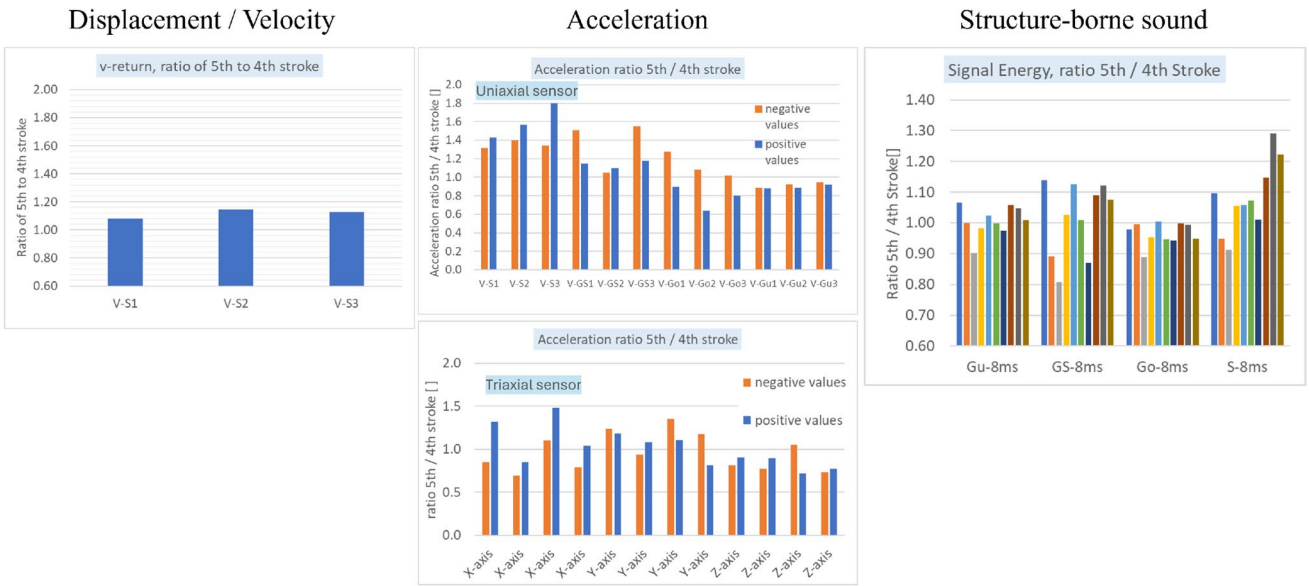


Fig. 22 Overview of the results on the problem of press blow intensity estimation

intend to improve the results by more sophistication data analysis tools, specifically machine learning methods in the future.

6 Conclusion

In this paper, we analyze the various sensor systems for the detection and intensity estimation of press blows in double-acting hammer systems. In practical laboratory tests and operational applications, three suitable methods and the necessary sensors are identified for the press blow identification. The press blows can be identified by evaluating the displacement/velocity signals of the hammer and/or by measuring values from acceleration or structure-borne sound sensors. The sensitivity of the acceleration and structure-borne sound measurements depends significantly on the position of the sensors on the various elements of the hammer unit. The measurement of the intensity of the rebound impacts required for controlling the hammer system is best captured by the acceleration (uniaxial) measurement method with the sensor position "Scabotte." The other two measurement methods are less suitable for estimating the intensity of the press blow.

Future work is concerned with the improved analysis of the obtained measurements using sophisticated signal analysis and machine learning methods. While the obtained results indicate the detectability, we expect improved performance for press blow intensity estimation. Further we intend to integrate the above sensor systems into control loops for the hammer with the aim to improve performance and energy consumption.

Appendix A Additional acceleration measurement results 1



Fig. 23 Comparison of the acceleration measurement results for a 12 fold loading using the uniaxial acceleration sensor

Appendix B Additional acceleration measurement results 2



Fig. 24 Comparison of the acceleration measurement results for a 12 fold loading using the triaxial acceleration sensor

Acknowledgements We thank Harald Hermanns for executing the project as well as for carrying out and documenting the experiments during his affiliation as a research associate at our lab.

Author Contributions Abdulkerim Karaman: investigation, data curation, visualization, figure preparation and translation, writing—review & editing; Andreas Schwung: conceptualization, methodology, formal analysis, writing—original draft, writing—review & editing; Michael Marre: supervision, resources, writing—review & editing. All authors read and approved the final manuscript.

Funding Open Access funding enabled and organized by Projekt DEAL. This work was funding by the Deutsche Bundesstiftung Umwelt (DBU; grant 33408/01). This Article is funded by the Open Access Publication Fund of South Westphalia University of Applied Sciences.

Data Availability Not applicable. All data supporting the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest/Competing interests The authors declare no competing interests.

Ethics approval and consent to participate This article does not contain any studies with human participants or animals performed by any of the authors.

Consent for publication Not applicable.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

1. Groover MP (2010) Fundamentals of Modern Manufacturing: Materials, Processes, and Systems. John Wiley & Sons, ???. https://books.google.de/books?id=QU-Qvud3OvoC
2. Altan T, Ngaile G, Shen G (2005) Cold and Hot Forging: Fundamentals and Applications. ASM International, ???
3. Zhbankov IG, Markov O, Perig AV (2014) Rational parameters of profiled workpieces for an upsetting process. Intern J Adv Manuf Technol 72:865–872
4. Uribe D, Durand C, Baudouin C, Bigot R (2024) Accurate real-time modeling for multiple-blow forging. Intern J Mater Form 17(57)
5. Behrens B, Bouguecha A, Lüken I, Mielke J, Bistron M (2014) Tribology in hot forging. Comprehens Mater Process 5:211–234
6. Markov O (2012) Forging of large pieces by tapered faces. Steel in Translation 42:808–810
7. Markov O, Zlygoriev V, Gerasimenko O, Hrudkina N, Shevtsov S (2018) Improving the quality of forgings based on upsetting the workpieces with concave facets. Eastern-European J Enterprise Technol 5(1):16–24
8. Markov O, Perig AV, Markova MA, Zlygoriev VN (2016) Development of a new process for forging plates using intensive plastic deformation. Intern J Adv Manuf Technol 83:2159–2174
9. Kim Y, Cho J, Bae W (2011) Efficient forging process to improve the closing effect of the inner void on an ultra-large ingot. J Mater Process Technol 211(6):1005–1013
10. Markov O, Gerasimenko O, Khvashchynskyi A, Zhytnikov R, Puzyr R (2019) Modeling the techological process of pipe forging without a mandrel. Eastern-European J Enterprise Technol 3(1 (99)):42–48
11. Kitayama S, Sugita R, Kondo S, Miyoshi K, Aono E, Amano M (2024) Numerical and experimental investigation of process parameters optimization in hammer forging for minimizing risk of crack. J Adv Mechan Design Syst Manuf 18:0087–0087. https://doi.org/10.1299/jamdsm.2024jamdsm0087
12. Sugita R, Kitayama S, Aono E, Kondo S, Miyoshi K, Amano M (2023) Numerical process parameters optimization in hammer forging for minimizing forging load. Proceed Intern Conf Design Concurr Eng Manuf Syst Conf 2023:7. https://doi.org/10.1299/jsmeidecon.2023.0_7
13. Saberi S, Fischer J, Stockinger M, Tikal R, Afsharnia R (2021) Theoretical and experimental investigations of mechanical vibrations of hot hammer forging. Intern J Adv Manuf Technol 114:3037–3045
14. Galdos L, Argandoña E, Herrero N, Ongay M, Adanez J, Sanchez M (2014) The calibration of high energy-rate impact forging hammers by the copper-column upsetting method and high speed camera measurements. In: Material Forming ESAFORM 2014. Key Engineering Materials, vol 611, pp 173–177. Trans Tech Publications Ltd, ???
15. Chen R, Li Z, Zhong K, Liu X, Wu Y, Wang C, Shi Y (2019) A stereo-vision system for measuring the ram speed of steam hammers in an environment with a large field of view and strong vibrations. Sensors 19(5)
16. Yoneyama T, Miyoshi K, Tate T (2018) Contact stress and temperature during air-stamp hammer upsetting of a circular cylinder. J Manuf Process 32:791–801
17. Markov O, Oleshko M, Mishina V (2011) Development of energy- saving technological process of shafts forging weighing more than 100 tons without ingot upsetting. Metallurg Mining Industry 3(7):87–90
18. Dindorf R, Wos P (2020) Energy-saving hot open die forging process of heavy steel forgings on an industrial hydraulic forging press. Energies 13(7)
19. Yan S, Xu J (2010) Study on dynamic characteristics of a hydraulic hammer. In: 2010 International conference on digital manufacturing & automation, vol 2, pp 459–462
20. Zhang YJ (2013) Research of all-hydraulic die forging hammer control system. In: Current Trends in the Development of Industry. Adv Mater Res 785:1286–1289. Trans Tech Publications Ltd, ???
21. Mull J-F, Durand C, Baudouin C, Bigot R (2020) A new tailored solution to predict blow efficiency and energy consumption of hammer-forging machines. Intern J Adv Manuf Technol 111:1941–1954
22. Uribe D, Durand C, Baudouin C, Bigot R (2025) Real-time forging process control: integrating billet-related surrogate and machine behavior models. J Intell Manuf

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.